

Ultimate Nutrition®
presents

LEUCINE

Amino acids form chains that become the proteins central to the development of muscle tissue, skin, and bone. They also play an important part in the bodily functions ancillary to muscle building: energy production, post-workout recovery, fat loss, healthy digestion, mood, and brain function. No chemical reaction takes place within the body without the aid of amino acids.

Overall, the body utilizes 22 standard amino acids for protein. The human body can synthesize 13 of these amino acids (non-essential amino acids; NEAAs) on its own and, therefore, doesn't require outside sources, such as food or supplements, to obtain them. The remaining 9 amino acids are classified as "essential" (or "indispensable") because they can only be obtained by external sources like food or supplements.

A closer look at the 9 EAAs shows three are classified as branched chain amino acids (BCAAs) based on their molecular structure: leucine, isoleucine, and valine. BCAAs are the most effective EAAs at reducing existing muscle breakdown and building new muscle after intense exercise. Of all the amino acids that help develop muscle tissue, the three BCAAs are responsible for as much as 35% of this production. Additionally, the presence of high amounts of BCAAs in the bloodstream during intense exercise slows the breakdown of energy (catabolism) and boosts the creation of energy (anabolism) during recovery.

During strenuous exercise, leucine is the BCAA that depletes most rapidly, thus making it the most crucial to replenish. Specifically, blood leucine levels decline by about 30% during both strength training and cardio workouts. Supplementing with leucine to increase availability has several benefits related to muscle protein synthesis (MPS). Leucine activates the main muscle building pathway, mTOR, for up to 24 hours post-workout. Furthermore, as people age, more leucine is typically needed to keep the mTOR pathway activated. In addition to signaling MPS, this pathway also triggers the brain to release transmitters that blunt hunger and extend satiety. Lastly, increased leucine levels in muscle tissue slow post-workout muscle breakdown caused by the degrading effects of cortisol, a stress hormone.

Each 1-scoop serving of Ultimate Nutrition's **Leucine** features 5 grams of rapid absorbing, free-form L-leucine. Free-form L-leucine is advantageous in that it doesn't take as long to digest as leucine derived from food (where the ingested protein first needs to be broken down into individual amino acids before the body can use it). This enables free-form L-leucine to pass quickly through the stomach and intestine and directly into the bloodstream for faster absorption. Add a scoop to your favorite protein shake or sports drink to replenish diminished leucine reserves and enhance muscle growth and recovery!

References

1. Churchward-Venne TA, Burd NA, Phillips SM. Nutritional regulation of muscle protein synthesis with resistance exercise: strategies to enhance anabolism. *Nutr Metab (Lond)*. May 17, 2012;9(1):40.
2. Churchward-Venne TA, Burd NA, Mitchell CJ, West DW, Philp A, Marcotte GR, Baker SK, Baar K, Phillips SM. Supplementation of a suboptimal protein dose with leucine or essential amino acids: effects on myofibrillar protein synthesis at rest and following resistance exercise in men. *J Physiol*. June 1, 2012;590:2751–65.
3. Mero AA, Hulmi JJ, Salmijärvi H, Katajavuori M, Haverinen M, Holviala J, Ridanpää T, Häkkinen K, Kovanen V, Ahtiainen JP, Selänne H. Resistance training induced increase in muscle fiber size in young and older men. *Eur J Appl Physiol*. March 2013. 113(3):641–50.
4. Loenneke JP, Wilson JM, Manninen AH, Wray ME, Barnes JT, Pujol TJ. Quality protein intake is inversely related with abdominal fat. *Nutr Metab (Lond)*. January 27, 2012;9(1):5
5. Thomson JS, Ali A, Rowlands DS. Leucine-protein supplemented recovery feeding enhances subsequent cycling performance in well-trained men. *Appl Physiol Nutr Metab*. April 2011;36(2):242–253.

These statements have not been evaluated by the Food and Drug Administration. This product is not intended to diagnose, treat, cure, or prevent any disease.